

Problem 1 (e):

The smallest model obtained in part (b) was the full integer INT8 quantized model, which achieved a size of 5.76 KB with 100% classification accuracy.

In part (e), I attempted to further reduce the model size by combining pruning and INT8 quantization. The idea was that pruning would remove less important weights while INT8 quantization would reduce storage precision.

However, the resulting pruned + INT8 model did not significantly reduce the TFLite model size beyond the original INT8 model. The classification performance remained high, but the storage size improvement was minimal.

This occurs because standard TFLite conversion does not fully compress sparse zero-valued weights created by pruning. As a result, most of the model size reduction already came from INT8 quantization alone.

Therefore, for this wine classification task, further size reduction beyond the INT8 model was difficult without significantly reducing the network architecture or risking performance degradation.

Problem 2:

1. Leaning blocks: Example

- a. Supervised Classification: Image classification CNN
- b. Supervised Regression: Huidility prediction model
- c. Anomaly Detection / Clustering: Vibration anomaly detection
- d. Transfer Learning: MobileNet image classifier
- e. Custom Learning Blocks: Custom neural network

2. Classifier Layers: Purpose

- a. Dense (Fully Connected): Performs classification or regression using learned weights
- b. Convolution (Conv1D / Conv2D): Extracts local spatial or temporal features
- c. MaxPooling: Reduces feature dimensions while preserving important features
- d. Flatten: Converts multidimensional tensors into vectors
- e. Dropout: Reduces overfitting by randomly disabling neurons during training
- f. Batch Normalization: Stabilizes and accelerates training
- g. Activation Layers (ReLU, Softmax, etc.): Introduces nonlinearity and probability outputs
- h. Reshape: Changes tensor dimensions between layers

3. Model Compression Options:
 - a. INT8 Quantization
 - b. Float16 Quantization
 - c. Dynamic Range Quantization
 - d. EON Compiler Optimization
4. Synthetic Data Generation
 - a. Audio: Whisper-based keyword spotting generation
 - b. Physics Simulation Data: Physics simulation dataset generation
 - c. 3D / Industrial Vision: NVIDIA Omniverse synthetic dataset generation
 - d. Images: DALL·E image generation